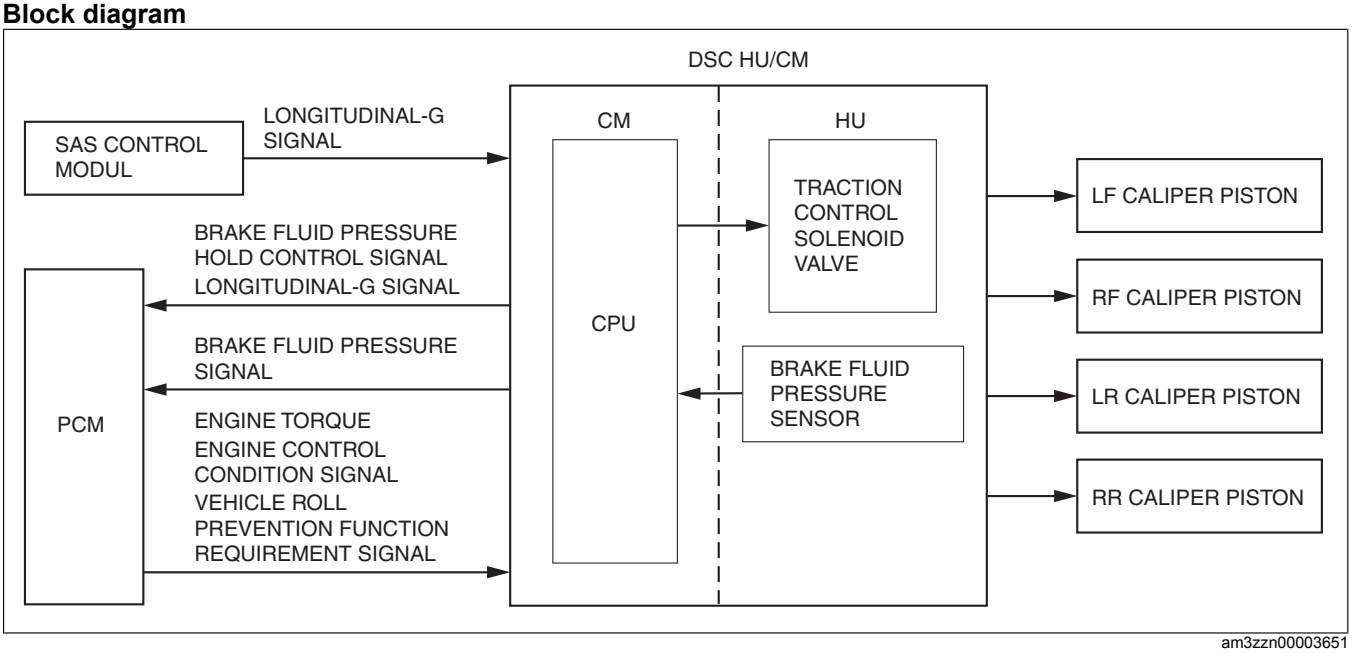
VEHICLE ROLL PREVENTION FUNCTION

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Note
- The vehicle roll prevention is a control available only on ATX vehicles. (Vehicles with i-stop)

Purpose/Function
- The vehicle roll prevention function operates the traction control solenoid valve in the DSC HU/CM to hold/reduce brake fluid pressure during i-stop control, and controls brake fluid pressure during i-stop control.
- The vehicle roll prevention function controls optimum brake fluid pressure based on each signal sent from the PCM.
  - — Maintains brake fluid pressure to maintain vehicle stop condition while i-stop (engine-stop control) is operating.
  - — Decreases brake fluid pressure according to road slope and engine torque for smooth startability while i-stop (engine-restart control) is operating.
- For details on i-stop, refer to the “i-stop CONTROL”. (See i-stop CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See i-stop CONTROL [SKYACTIV-D 1.5].)

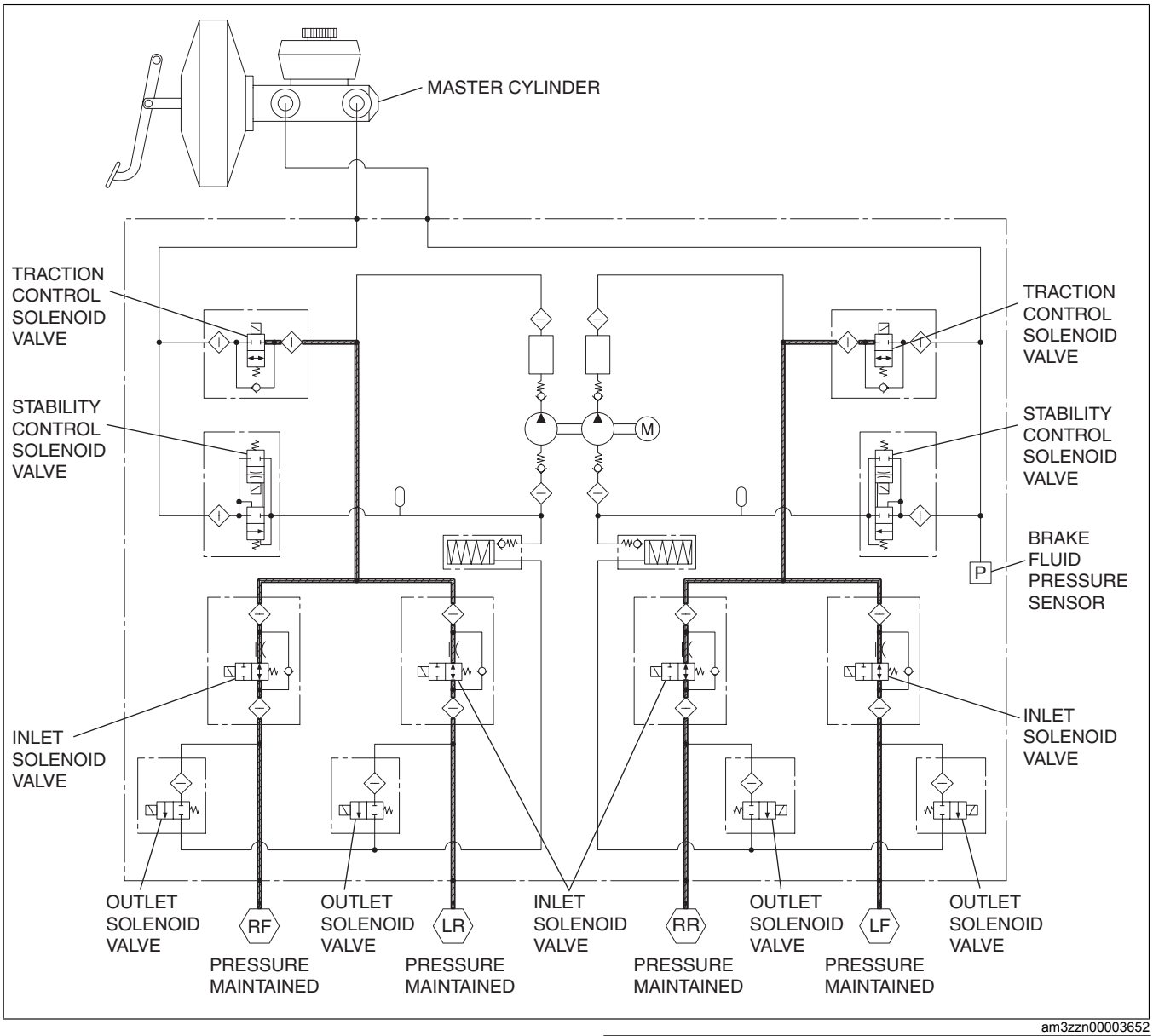
Construction
Block diagram



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Operation
- The vehicle roll prevention function controls the traction control solenoid valve in the DSC HU/CM to maintain or decrease brake fluid pressure by the following signals sent from the PCM and longitudinal-G signal sent from the SAS control module.
  - — Vehicle roll prevention function control request signal
  - — Engine control condition signal
  - — Engine torque signal
- The vehicle roll prevention function controls as follows according to the i-stop control condition:
  i-stop control (engine stop)
  - — When the DSC HU/CM receives the vehicle roll prevention function control request signal from the PCM, controls the brake fluid pressure hold control (traction control solenoid valve is energized and hydraulic circuit is closed) and maintains the brake fluid pressure while the vehicle is stopped.

Hydraulic circuit diagram

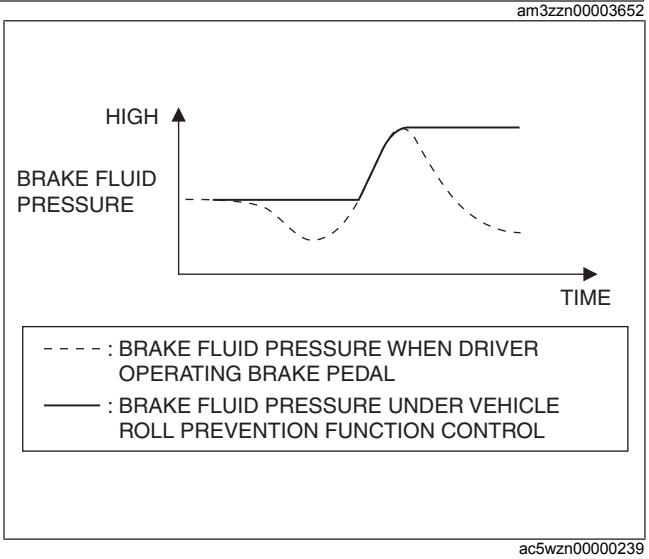


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— If the driver depresses the brake pedal and the brake fluid pressure increases, the increased brake fluid pressure is maintained.
— The DSC HU/CM sends hold control signal to the PCM.

i-stop control (engine restart)

— The DSC HU/CM operates brake fluid pressure release control (maintained fluid pressure is controlled by changing energization amount to traction control solenoid valve) according to the engine torque signal sent from the PCM and the longitudinal-G signal from the SAS control module.
— When the engine torque signal sent from the PCM exceeds the specification after the engine restarts, the brake fluid pressure decreases. The time for decreasing changes depending on road slope and the time for decreasing becomes longer than on level road. In addition, when the accelerator pedal is depressed, the time for decreasing becomes shorter.



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